

SKILL WEEK 7

Task 1:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{ pid_t
pid;
int status;

    pid = fork();
    if (pid < 0) {
        perror("fork failed");    return
        1;
    }

    if (pid == 0) {

        // Child process    printf("Child process
        started.\n");    printf("Child
        PID: %d\n", getpid());

        sleep(3);

        printf("Child process completed.\n");

        exit(0);

    } else {

        // Parent process    printf("Parent process
        waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {    printf("Child exited
        normally.\n");    printf("Child exit status: %d\n",
        WEXITSTATUS(status));
        }
    }
}
```

```
    printf("Parent process completed.\n");
}
```

```
return 0;
```

```
naveen123@LAPTOP-4NVQJ2C7:~$ nano skill7.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc skill7.c -o skill7
naveen123@LAPTOP-4NVQJ2C7:~$ ./skill7
Parent process waiting for child...
Child process started.
Child PID: 1134
Child process completed.
Child exited normally.
Child exit status: 0
Parent process completed.
naveen123@LAPTOP-4NVQJ2C7:~$ |
```

Task 2:

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main() {

    char *path = getenv("PATH");

    if (path == NULL) {        printf("PATH
variable not found.\n");      return 1;
    }

    printf("PATH:\n%s\n\n", path);

    char *copy = strdup(path);
    char *directory = strtok(copy, ":");

    char filename[100];

    printf("Enter executable name to search: ");
    scanf("%99s", filename);

    while (directory != NULL)

        { char fullpath[500];
```

```

        snprintf(fullpath, sizeof(fullpath),
                 "%s/%s", directory, filename);

        if (access(fullpath, X_OK) == 0)
        { printf("Executable found: %s\n", fullpath);
          free(copy);
        }
        return 0;
    }

    directory = strtok(NULL, ":");
}

printf("Executable not found in PATH.\n");

free(copy);

return 0;
}

```

Output:

```

naveen123@LAPTOP-4NVQJ2C7:~$ nano skill7.1.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc skill7.1.c -o skill7.1
naveen123@LAPTOP-4NVQJ2C7:~$ ./skill7.1
PATH:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/Windo
wsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/Nave
en/AppData/Local/Programs/Python/Python310/Scripts:/mnt/c/Users/Naveen/AppData/Local/Programs/Python/Python310:/mnt/c/Users/Naveen/AppData/Local/Programs/
Python/Python315/Scripts:/mnt/c/Users/Naveen/AppData/Local/Programs/Python/Python315:/mnt/c/Users/Naveen/AppData/Local/Programs/Python/Launcher:/mnt/c/Us
ers/Naveen/AppData/Local/Microsoft/WindowsApps:/mnt/c/cprog/S10C/bin:/mnt/c/Users/Naveen/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/J
ava/jdk-25/bin:/mnt/c/Users/Naveen/AppData/Roaming/npm:/mnt/c/Users/Naveen/AppData/Local/kingsoft/WPS Office/12.1.0.27458/clitool:/snap/bin

Enter executable name to search: ls
Executable found: /usr/bin/ls
naveen123@LAPTOP-4NVQJ2C7:~$

```